Winning and Attendance in the WNBA

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A central question in the intersection between sports economics and sports analytics is how to quantify the effects of what happens on the court or field on a team's revenue. In general, the literature expects that stronger teams will see higher live game attendance, but previous research has not found a statistically significant relationship between winning percentage and attendance in the WNBA (Agha and Berri 2023). Our study improves upon theirs by using a rolling win count from the past 44 games played rather than a seasonal winning-percentage to predict per game attendance. This avoids look-ahead bias, as fans cannot be expected to know a team's future performance when deciding to attend a game. Using game-level data for all WNBA regular-season games from 2018–2024, we estimated an ordinary least squares model of per-game attendance with team and season fixed effects and a dummy variable for whether Caitlin Clark played. We then translated player win shares into expected attendance impacts by mapping individual contributions to team wins onto the estimated attendance–winning relationship. We chose to focus on attendance as a proxy for revenue following the research framework from Scully (1974), due to concerns about revenue data availability. We used attendance data from the "Across the Timeline: WNBA League Attendance Dashboard", which records game-level attendance data for all WNBA games. We find a significant relationship between rolling win count and game attendance, with game attendance expected to increase by 112 people for each additional win in the 44 game window. This means that a player like Allisha Gray, for example, who we estimate contributed 10 wins to the Atlanta Dream in 2024, boosted attendance by approximately 1,100 attendees per game. These results highlight the importance of short-run performance in shaping fan demand and provide a framework for linking individual player performance to team-level revenue outcomes in women's professional sports.

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1 Introduction

1.1 Gabriela introduction

In the years prior to the 2020 WNBA collective bargaining agreement (CBA), NBA Commissioner Adam Silver publicly characterized the women's league as unpopular and unprofitable (Garcia 2020). Again during the CBA renegotiations in 2025, NBA and WNBA leadership made similar negative claims about the women's league's long term viability. While league leadership has been ironically pessimistic about the health of the WNBA in recent years, their claims stand in tension with recent trends: the league has enjoyed a significant increase in attendance and fan interest in the past several seasons.

While this trend is encouraging for anyone interested in the success of the WNBA, breaking down further the underlying components of this trend is necessary both to explain and sustain this growth. For decades, league officials and sports economists have debated the primary drivers of game attendance, identifying game-related and market-related factors impacting fan interest and game attendance. Much of this research, beginning with foundational work in the 1970s, assumes that fans are more likely to attend games when the home team is expected to do well. This relationship is widely tested and supported in men's professional sports, but women's leagues and the WNBA in particular are still largely underresearched. As a relatively young league with rapid growth in viewership (WNBA 2024) but limited publicly available financial data, game attendance remains the most readily available proxy variable for public interest in the WNBA.

A key line of research in sports economics links fan demand to expectations about game outcomes. Early work such as the uncertainty-of-outcome hypothesis suggests that attendance is higher when game results are less predictable, increasing perceived excitement. In their 2014 paper, Coates et al use more recent behavioral economics models to suggest that people evaluate gains and losses relative to a reference point and tend to dislike losses more than they like equivalent gains (Coates et al. 2014). They argue that applying this idea of loss aversion may better explain why fans choose to go games than the uncertainty of outcome hypothesis. They use MLB live-game attendance data to find attendance patterns consistent with the hypothesis that fans are less likely to attend games in which they expect the home-team to lose (Coates et al. 2014). These frameworks together suggest that attendance decisions are sensitive not only to team quality, but to expectations about competitive performance and likely game outcomes.

In the first study in sports economics that sought to empirically identify the factors that affected attendance, Noll (1974) considered data from MLB, the NFL, NHL, NBA and ABA (American Basketball Association) for the 1969-70 and 1970-71 seasons and found that demand was a function of population, interactions of income with star players, team win percentage, and per capita income. Multiple subsequent studies on men's professional team sports continued to find a statistically significant correlation between team quality and attendance. In these studies, a debate emerged about the direction of causality: did strong team performance engender higher attendance or vice versa (Baade and Tiehen 1990; Clapp and Hakes 2005; Davis 2008, 2009; Horowitz 2007; Lemke et al. 2010)? While no Granger causality tests were run, the predominant view has been causality running from team performance to attendance (Hall et al. 2002).

Finally, situating these revenue models within broader league structure, sports economists have written extensively about the relationship between winningness and team revenue, particularly about how competitive balance benefits teams across the league in generating revenue. While teams and fans want to win as many games as possible, significant imbalance in winning percentages across a professional sports league will tend to decrease viewership as spectators will only watch a game if they have some uncertainty about the outcome (Perline et al. 2018). In their study of competitive balance in NCAA basketball, Perline et al. found that the women's game was significantly less balanced than the men's game, looking at the standard deviation and the range of winning percentages for teams in the women's division I conference (Perline et al. 2018). They hypothesize that the difference in the competitive balance of the men's and women's games could be one cause of lower revenue generation and attendance in the women's game.

In their study of attendance determinants in the WNBA, Agha and Berri argue that a major reason why attendance is considerably lower at WNBA games than NBA games is that the women's league is still in its early years, only beginning play in 1997 compared to the NBA's 1946. To account for this, they compare attendance in the tenth season of the WNBA (2006) and the NBA (1955/56) (Agha and Berri, 2023, p. 36.) In their respective tenth years, the WNBA average attendance per home game exceeded the NBA's by more than 2,000 fans per game. This comparison suggests that the current disparity in attendance between the women's and men's league should not be interpreted as a lack of demand, but rather a sign of the league's maturity. Considering the league's age as a factor in demand for attendance, it is interesting to inquire further into whether patterns of fan behavior change as the league matures. Agha and Berri attempt to establish whether different forces shape demand for attendance in the WNBA and the NBA, finding that team performance was not a statistically significant predictor of game attendance in the WNBA. As noted above, this is contrary to the virtually unanimous finding in the existing scholarship for men's leagues. However, their study only employed data from 2004 through 2017 and thus can not capture the more recent improvements in attendance shown in figure 1.

Of course, all of this differential cannot be attributed to the age of the league. Additional important factors include: the WNBA plays during the summer months in order to promote arena use outside of the NBA season; there is less money spent on promotion, reduced media coverage, historical prejudice, arena amenities and age, and ticket prices, among others. To be sure, the sports management literature also includes several studies that find a major motivation behind fandom for women's sports leagues to be an area's overall support for feminist issues (Delia 2020; Guest and Luijten 2018; Leslie-Walker and Mulvenna 2022). Nonetheless, to the extent that the league's age is a factor, it is interesting to inquire into the patterns of fan behavior and whether they might change as the league matures. Agha and Berri (2023) found that their variable on team performance was not statistically significant as an explainer of team attendance. As noted above, this is contrary to heretofore virtually unanimous finding in the existing scholarship for men's leagues. Note, however, that Agha and Berri (2023) employed data only from 2004 through 2017 and that its performance variable was team win percentage during the previous year.

In viewing more recent attendance figures for the WNBA in the context of the league's history, we hoped to determine whether predictors of attendance have shifted over time, due to the WNBA's relative instability as a young league. Specifically, we find that in recent years the strength of the effect of team strength on game attendance has increased, paralleling documented trends in men's leagues.

In this paper we build upon the literature on team performance and game attendance in the context of the WNBA and introduce a period-based model that allows us to more accurately capture the current state of league growth.

1.2 still to be incorporated in intro:

Differences in the literature over how to measure team quality (wins, wins lagged, number of championships in past years, win probability, star power), attendance (logs, lags, linear regression or censored), the inclusion of control variables and non-linear forms also emerged (Wiseman and Chatterjee 2003; Woltring et al. 2018; Kim and Chang 2023; DeSchriver and Jensen 2002; Paul et al. 2019).

The use of only last year's win percentage to measure team performance is problematic because fans are more likely to respond to the more recent performance of the team. Thus, while a team's win percentage in 2016 might have an appreciable impact on the team's attendance at the beginning of the 2017 season, as the season wears on, this impact should consistently diminish.

Professional men's team sport leagues have demonstrated a clear relationship between team quality and attendance. It is reasonable to anticipate that as women's sports mature and are perceived increasingly as a normal sports league, rather than a political concession or oddity, they will begin to assert similar fan demand characteristics. In particular, fan demand will become more responsive to team quality. Using game attendance[1] and team performance data from 1997 through 2025, we test these hypotheses.

[1]Note that, as in most sports leagues, there is always the possibility finagling with game attendance figures. The WNBA appears to give attendance counting instructions to teams that define attendance as turnstile count, plus season ticket sales, plus complimentary tickets and promotions. There is considerable for these factors to be manipulated, rendering the possibility of noise in the attendance figures we use.

1.3 Andrew other passage

The relationship between attendance and winning percentage has been examined, but such studies often look at winning percentage as causation for attendance. Davis (2008) and Horowitz (2007) both found that attendance and winning percentage were related. Horowitz concluded a bidirectional relationship between the two, while Davis suggested that causation runs from winning percentage to attendance. In a following study, Davis (2009) found winning percentage as a significant determinant of attendance for 12 National League teams; though the study did not establish causation both ways. Lemke et al. (2010) found attendance increases as home team win probability increases. There is evidence that causation may run from attendance to winning percentage, but the bulk of the literature has been researched the other way.

The present study controls for payroll and stadium capacity as both have been shown to affect attendance. Wiseman and Chatterjee (2003) found team payroll was positively correlated to team winning percentage (9). Additionally, they found that this relationship has grown stronger since the late 1990's. Hall et al. (2002) did not find causality evidence from payroll to team winning percentage in their entire dataset, but they did note an increased correlation in the 1990's. Additionally, they stated that an MLB team with a payroll of at least 150% of the league average could expect a winning percentage of at least .550. This is an important finding, because Vrooman (2012) identified .550 as a benchmark winning percentage for qualifying as a playoff team.

2 Data

2.1 Game logs and attendance

Our analysis is based on game-level data that spans WNBA history, beginning in 1997 and continuing through the end of the 2025 season. Our final data set contains observations from 6,164 games.

First, we collected game-level attendance data from (Across the Timeline, n.d.). We supplemented these data with arena capacities that we collected manually[2]. Attendance data was unavailable for only 152 games, while (due to the COVID-19 pandemic) 2 games (accurately) recorded attendance of 0.

Second, we collected game-level performance data from the **wehoop** R package (Gilani and Hutchinson 2021), which uses the official WNBA API. These data include the final score of the game. We also used the **wehoop** package to identify whether Caitlin Clark played more than 0 minutes in the game.

[2] We were only able to collect an estimate for the *current* arena capacity, which is not necessarily the same as the arena capacity on the day of the game.

Table 1

WNBA Game Data (1997-2025)

Sample of game-level records

Game Date	Home	Away	Arena	Attendance	Plus Minus	Clark
19596	SEA	CHI	Climate Pledge Arena	9,893	-5	0
15492	SAN	PHO	AT&T Center	6,534	19	0
18068	CHI	IND	Wintrust Arena	4,945	-7	0
13291	SEA	HOU	KeyArena	7,541	-25	0
13732	LAS	PHO	Staples Center	10,079	-3	0
19868	PHO	DAL	Footprint Center	8,339	-15	0
11895	PHO	SAC	US Airways Center	9,344	-15	0
12258	SEA	PHO	KeyArena	6,753	29	0
16591	CHI	IND	Allstate Arena	8,123	23	0
18042	LVA	LAS	Mandalay Bay Events Center	7,249	13	0

Sample of Game-level WNBA data. The data indicates (among other things) which teams were playing, who won, the arena, the attendance, and whether Caitlin Clark played in the game.

Table 1 shows a sample of the data. While many other variables are omitted from the table, we note that the attendance, the teams playing in the game, the outcome, the arena, and whether Caitlin Clark played in the game, are recorded.

Source: Article Notebook

Figure 1 shows the distribution of attendance across all seasons, with a logarithmic scale on the vertical axis. There is no attendance data from 2020, because all games that year were played in the WNBA bubble due to the COVID-19 pandemic, and some games in 2021 continued to be played with artificially low attendance.

Source: Article Notebook

Several broad observations can be made from Figure 1. First, after some initial enthusiasm during the first three years of the league, attendance largely plateaued or steadily but slightly declined from 2000 to 2017. The two years preceding the pandemic (2018-2019) had notably lower attendance. Since the pandemic, attendance has rebounded, exceeding its initial levels in 2024 and reaching new heights in 2025. In fact, average WNBA attendance in 2025 was 10,986, which was 1.1% higher than the previous high in 1998.

These last two years coincide with the arrival of Caitlin Clark, the 2024 Rookie of the Year Award winner who is widely regarded as a generational talent. Disentangling the “Caitlin Clark effect” from the league’s overall attendance trends is one of our goals.

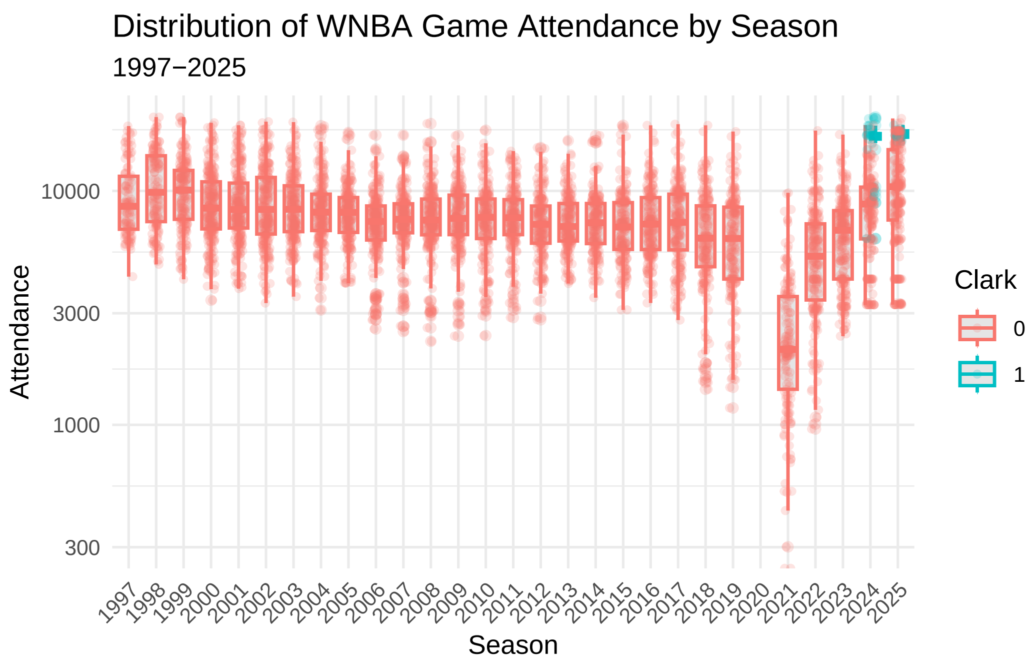

Figure 1: Distribution of WNBA game attendance by season. Note that the vertical axis has a logarithmic scale. Caitlin Clark's games in 2024-2025 are separated by color.

Source: Article Notebook

2.2 Estimating team strength

We used the game-level results data to model the strength of each WNBA team as a time series. Measuring team strength is necessary to test our hypothesis that attendance is in part driven by the desire of fans to see quality basketball. We considered season-to-date winning percentage and a rolling 44-game (the length of a typical WNBA season) winning percentage before settling on Elo rating (Elo 1978). Originally developed for rating chess players, Elo rating is well-understood in sports analytics as a measure of team strength that updates after each game, and—unlike winning percentage and rolling winning percentage—takes into account not only the outcome but also the estimated team strength of both teams.

Let $\theta_i(t)$ be the strength of team i at time t, where Elo rating are typically centered at 1500. Then if the probability that team i beats team j is modeled by a logistic curve with base 10 and scale factor 400, the win probability is given by:

$$\Pr(i \text{ beats } j \text{ at time } t) = \frac{1}{1 + 10^{\frac{\theta_j(t) - \theta_i(t)}{400}}}\,.$$

Each team starts the 1997 season with $\hat{\theta}_i(0) = 1500$, and following each game, team i's rating is updated via the formula

$$\hat{\theta}_i(t+1) = \hat{\theta}_i(t) + K \cdot I(i \text{ beats } j)\,,$$

where $I(i \text{ beats } j)$ is an indicator variable that is 1 if i beats j and -1 if j beats i, and $K = 32$ is a constant that controls how sensitive the ratings are to each game. Figure 2 shows the evolution of Elo rating $\hat{\theta}_i(t)$ by franchise for the 12 teams who were active during the 2024 season. The recent poor performance of the Los Angeles Sparks, the 2010 dominance of the Seattle Storm, and the recent rise of the New York Liberty and Indiana Fever are easily visible.

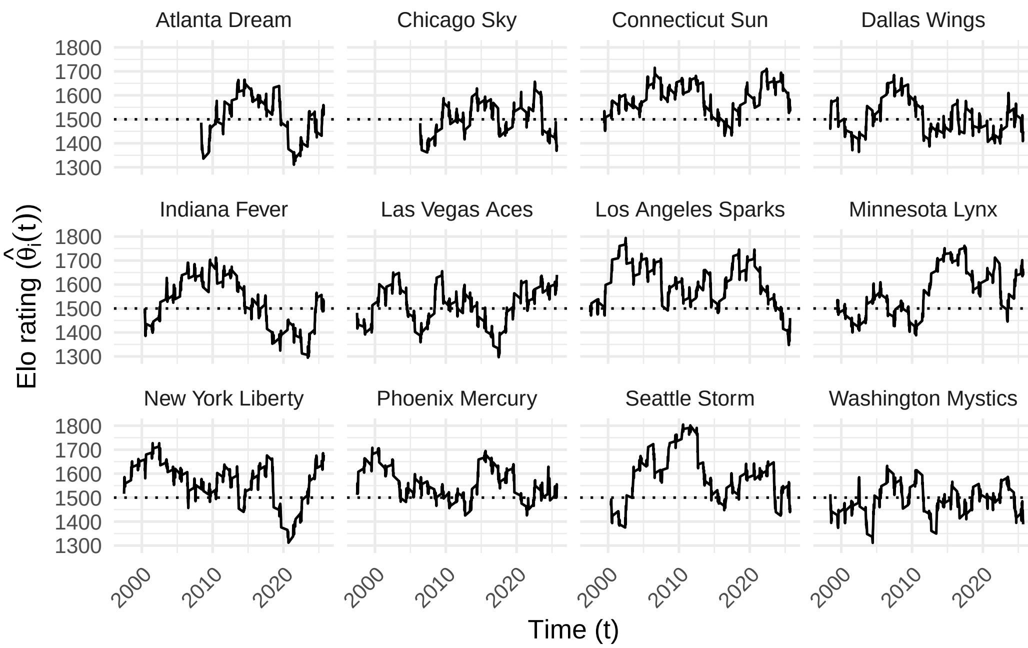

Figure 2: Evolution of team Elo ratings in the WNBA (1997-2025). Only franchises active in 2024 are shown. The recent poor performance of the Los Angeles Sparks, the 2010 dominance of the Seattle Storm, and the recent rise of the New York Liberty and Indiana Fever are easily visible. Note that all franchises begin with a 1500 rating.

Source: Article Notebook

2.3 Estimating home win probability

The Elo ratings estimated above can then be plugged back into the win probability formula to estimate the home teams's chances of winning. We also added a 12-point home court advantage

Table 2: Calibration of home win probability model, based on Elo ratings. We note that in the most frequent situations, where the home team was a slight favorite, the observed home win probability is within 3 percentage points of the predicted home probability. However, in those rarer occasions where a home win was extremely likely or unlikely, the model was less accurate.

WNBA Home Win Probability Model Calibration (1997-2025)

10 Equally-spaced Bins

Wpct Bin	Games	Bin Midpoint	Obs Home Wpct	Diff
(0.153,0.233]	51	0.212	0.059	-0.153
(0.233,0.311]	170	0.284	0.188	-0.096
(0.311,0.39]	441	0.360	0.211	-0.149
(0.39,0.468]	744	0.432	0.349	-0.083
(0.468,0.547]	971	0.508	0.485	-0.023
(0.547,0.625]	1,136	0.586	0.601	0.016
(0.625,0.704]	1,150	0.664	0.697	0.033
(0.704,0.782]	910	0.737	0.815	0.078
(0.782,0.861]	490	0.810	0.894	0.084
(0.861,0.94]	101	0.884	0.950	0.066

to our home win probability model, so that:

$$\hat{p}_{ij}(\hat{\theta}_i(t), \hat{\theta}_j(t), t) = \Pr(i \text{ beats } j \text{ at time } t \text{ at home}) = \frac{1}{1 + 10^{\frac{\hat{\theta}_j(t) - (\theta_i(t) + 12)}{400}}}.$$

We arrived at 12 points by manually calibrating the win probabilities to match the observed frequencies. While the model was less accurate at the extremes (it was consistently reluctant to predict nearly certain home wins and losses), it fit very well in the middle where most of the games occur (see Table 2).

Source: Article Notebook

3 Methods

Our goal is to model the attendance (y) at a given WNBA game, as a function of several different measures of fan engagement detailed in Section 3.2. We also include several control variables to account for time and city-related effects (see Section 3.3).

We also know the arena in which the game took place and the capacity of that arena. Thus, and alternative response variable is percentage of arena capacity filled (γ).

3.1 Response variables

In Figure 3, we observe the relationship between attendance and estimated home win probability. In most years, the slope appears to be positive.

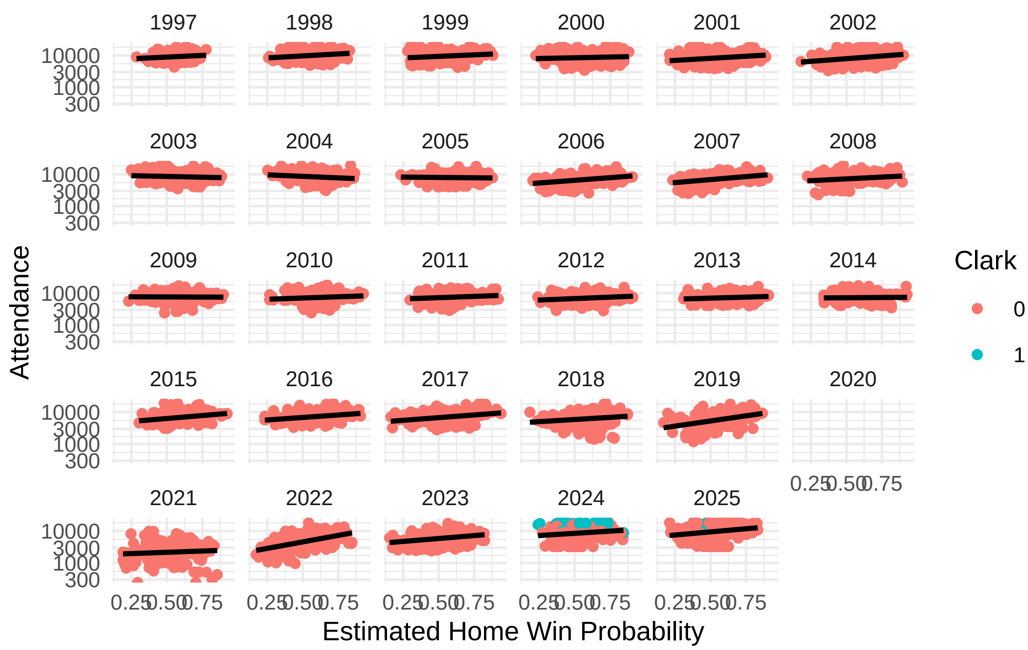

Figure 3: WNBA attendance by season, against estimated home win probability. We note that in most years, the slope of the association appears to be positive.

Source: Article Notebook

In Figure 4, we note similar trends in the relationship between the percentage of the arena capacity filled and the estimated home win probability. We also note that since arena capacity cannot exceed 1, there are edge effects created by sellouts. Those sellouts are much more frequent in the last two years of data, and many of the sellouts involve Caitlin Clark.

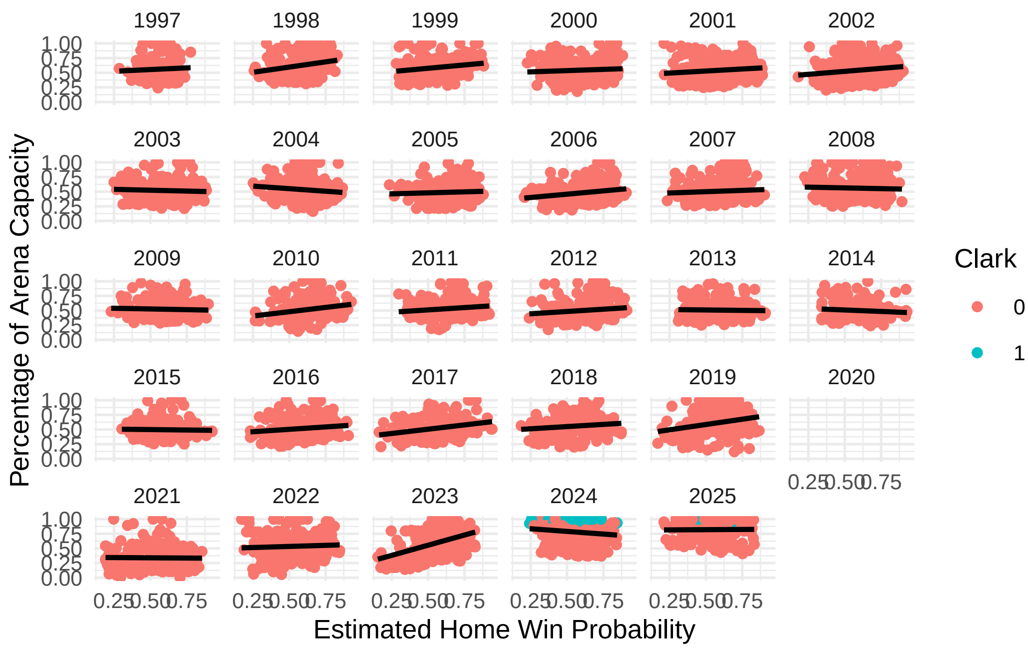

Figure 4: WNBA percentage of arena capacity filled by season, against estimated home win probability. We note the edge effects caused by sellouts, which occur much more frequently in 2024 and 2025.

Source: Article Notebook

Because the raw attendance figures are not normally distributed, we model the logarithm of attendance. Similarly, because the percentage of arena capacity are bound above by 1 and below by 0, we employ a Tobit regression model.

3.2 Modeling fan engagement

In Section 2.2, we described our use of Elo rating to estimate the strength of the home team ($\hat{\theta}_i(t)$), the strength of the visiting team ($\hat{\theta}_j(t)$) and the home win probability ($\hat{p}_{ij}(t)$) (which is itself a function of the estimated team strengths).

As noted in Section 1, there are multiple notions of what drives fan engagement relevant to the WNBA. Each of these notions can be captured by a simple bivariate function of the corresponding strengths of the teams ($f_k(\hat{\theta}_i(t), \hat{\theta}_i(t))$).

- Quality of play: fans may wish to attend a game in person if the quality of play is high. We model this as the sum of the Elo rating of the two teams ($f_1(\hat{\theta}_i(t), \hat{\theta}_i(t)) = \hat{\theta}_i(t) + \hat{\theta}_j(t)$)

- Loss aversion: fans may wish to attend a game to see their team (the home team) win. We model this using the estimated home win probability function above ($f_2(\hat{\theta}_i(t), \hat{\theta}_i(t)) = \hat{p}_{ij}(t)$)
- Uncertainty outcome hypothesis: fans may wish to attend a game for which the outcome is unclear. We model this as ($f_3(\hat{\theta}_i(t), \hat{\theta}_i(t)) = |\hat{\theta}_i(t) - \hat{\theta}_j(t)|$)

3.3 Control variables

As time-related control variables, we include the the season (τ_1), the month of the year (τ_2), and day of the week (τ_3) as fixed effects. To control for city-related effects, we also include the team name (ϕ_1), as well as the median income (ϕ_2, in thousands of USD) and population (ϕ_3, in millions) of the home city[3]. Finally, we include an indicator based on whether Caitlin Clark played more than 0 minutes in the game (χ).

3.4 Interaction

Testing our primary hypotheses requires analyzing how the relationship between attendance and fan engagement has changed over time. To capture this, we include an interaction term between our measures of fan engagement (f_k) and season (τ_1).

3.5 Model formulation for attendance

Like Coates et al. (2014), we model the natural logarithm of attendance.

Our general regression model for attendance is:

$$\ln y = \beta_0^{(k)} + \beta_1^{(k)} f_k + \underbrace{\beta_2^{(k)}\tau_1 + \beta_3^{(k)}\tau_2 + \beta_4^{(k)}\tau_3}_{\text{time-related}} + \underbrace{\beta_5^{(k)}\phi_1 + \beta_6^{(k)}\phi_2 + \beta_7^{(k)}\phi_3}_{\text{city-related}} + \beta_8^{(k)}\chi + \underbrace{\beta_9^{(k)} f_k \tau_1}_{\text{interaction}} + \epsilon\,,$$

where $k \in \{1, 2, 3\}$, and $\beta_j^{(k)} \in \mathbb{R}$ for $j = \{0, 1, 6, 7, 8\}$, while the other $\beta_j^{(k)}$ coefficients are vectors of length equal to the number of factor levels in their corresponding categorical explanatory variable. Most notably $\beta_2^{(k)}$ and $\beta_9^{(k)}$ have 29 values (corresponding to each season after 1997) and $\beta_5^{(k)}$ has 19 values (corresponding to each team). Only current WNBA franchises are included in the regression model. We also excluded all data from the 2020 and 2021 seasons, since there were no fans allowed in all or parts of both seasons.

[3]Because the population and median income statistics were available at irregular time periods (due to the COVID-19 pandemic and other factors), we chose to use the median of the population and median income across all of the available years of data as a constant value. While this does mean that the growth in cities like Phoenix or Las Vegas is ignored, we doubt this will meaningfully affect our results, especially since the team fixed effect should capture some of that variability.

3.6 Model formulation for percentage of arena capacity

We fit the same set of models with γ as the response, but the in Tobit framework.

4 Results

Source: Article Notebook

4.1 Attendance

In keeping with recent guidance from Wasserstein and Lazar (2016) and Wasserstein et al. (2019), and with a sample size of over 5000 observations and dozens of coefficients, we are less interested in the p-values shown in Table 3, and focus our attention on the magnitude and practical impact of the fitted coefficients. (All variables except for population achieve "statistical significance" at the 5%-level, as shown in Table 4). Residual analysis for the attendance models are shown in Section 6.1.

Table 3: Fitted coefficients for scalar terms in models for the natural logarithm of attendance.

Model	Term	Estimate	Std Error	Statistic	P Value
f1	(Intercept)	9.027	0.129	69.908	0.000
f2	(Intercept)	8.968	0.155	58.003	0.000
f3	(Intercept)	9.078	0.132	68.750	0.000
f2	I(abs(elo.A - elo.B))	0.001	0.001	0.524	0.600
f1	I(elo.A + elo.B - 3000)	0.002	0.001	3.320	0.001
f3	I(home_win_prob - 0.5)	0.422	0.402	1.050	0.294
f1	is_cc	0.705	0.049	14.514	0.000
f2	is_cc	0.699	0.050	14.089	0.000
f3	is_cc	0.653	0.048	13.627	0.000
f1	median_income	-0.007	0.002	-3.687	0.000
f2	median_income	-0.006	0.002	-3.144	0.002
f3	median_income	-0.008	0.002	-4.003	0.000
f1	median_pop	0.021	0.007	2.967	0.003
f2	median_pop	0.014	0.007	1.888	0.059
f3	median_pop	0.017	0.007	2.382	0.017

Source: Article Notebook

Consider first the strong evidence for the Caitlin Clark effect. Even for the smallest coefficient in Table 3, the expected attendance nearly doubles (i.e., is multiplied by $\exp(0.653) = 1.92$)

when Clark plays in the game, and this is *after* controlling for the Indiana Fever team effect, the level of fan engagement, and the season effects!

Conversely, the evidence for the quality of play and uncertainty of outcome hypotheses is weak, with those coefficients barely moving the needle. A 10 percentage point increase in the probability of the home team winning is associated with a modest 4.31% increase in the expected attendance, after controlling for our other factors.

Table 4: Analysis of variance for terms in models for the natural logarithm of attendance.

Model	Term	Df	Sumsq	Meansq	Statistic	P Value
f2	I(abs(elo.A - elo.B))	1	10.851	10.851	99.987	0.000
f1	I(elo.A + elo.B - 3000)	1	40.473	40.473	401.255	0.000
f3	I(home_win_prob - 0.5)	1	39.923	39.923	395.845	0.000
f1	is_cc	1	22.763	22.763	225.675	0.000
f2	is_cc	1	21.472	21.472	197.854	0.000
f3	is_cc	1	19.936	19.936	197.668	0.000
f1	median_income	1	1.502	1.502	14.890	0.000
f2	median_income	1	1.307	1.307	12.039	0.001
f3	median_income	1	1.629	1.629	16.153	0.000
f1	median_pop	1	0.278	0.278	2.759	0.097
f2	median_pop	1	0.322	0.322	2.964	0.085
f3	median_pop	1	0.154	0.154	1.528	0.216
f1	factor(month(game_date))	4	12.644	3.161	31.339	0.000
f2	factor(month(game_date))	4	12.434	3.109	28.645	0.000
f3	factor(month(game_date))	4	11.842	2.960	29.354	0.000
f1	factor(wday(game_date))	6	14.323	2.387	23.667	0.000
f2	factor(wday(game_date))	6	15.293	2.549	23.487	0.000
f3	factor(wday(game_date))	6	14.704	2.451	24.300	0.000
f1	name_current	12	117.498	9.791	97.074	0.000
f2	name_current	12	132.317	11.026	101.605	0.000
f3	name_current	12	122.443	10.204	101.171	0.000
f2	I(abs(elo.A - elo.B)):factor(season_id)	26	6.526	0.251	2.313	0.000
f1	I(elo.A + elo.B - 3000):factor(season_id)	26	27.457	1.056	10.470	0.000
f3	I(home_win_prob - 0.5):factor(season_id)	26	31.440	1.209	11.990	0.000
f1	factor(season_id)	26	152.181	5.853	58.029	0.000
f2	factor(season_id)	26	150.433	5.786	53.315	0.000
f3	factor(season_id)	26	147.101	5.658	56.098	0.000

Source: Article Notebook

Table 5 characterizes the model fits. We can see that while all models achieve a similar fit, models f_1 (corresponding to the quality of play hypothesis) and f_3 (corresponding to the home win probability) are slightly superior by most metrics.

Table 5: Comparison of models.

model	r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual	nobs
f1	0.436	0.427	0.318	49.459	0	78	-1337.102	2834.204	3356.582	502.715	4984	5063
f2	0.394	0.384	0.329	41.460	0	78	-1522.340	3204.680	3727.057	540.879	4984	5063
f3	0.436	0.428	0.318	49.471	0	78	-1336.828	2833.656	3356.033	502.660	4984	5063

Source: Article Notebook

Figure 5 shows that after controlling for the Caitlin Clark effect, expected attendance is only just returning its 1997 baseline in the most recent season. This is a mixed blessing. On the one hand, even after controlling for Clark, league expected attendance has fully recovered from the COVID-19 pandemic and the malaise of the early 2000s and 2010s. On the other hand, it seems unlikely that the league would have achieved record high attendance without Clark.

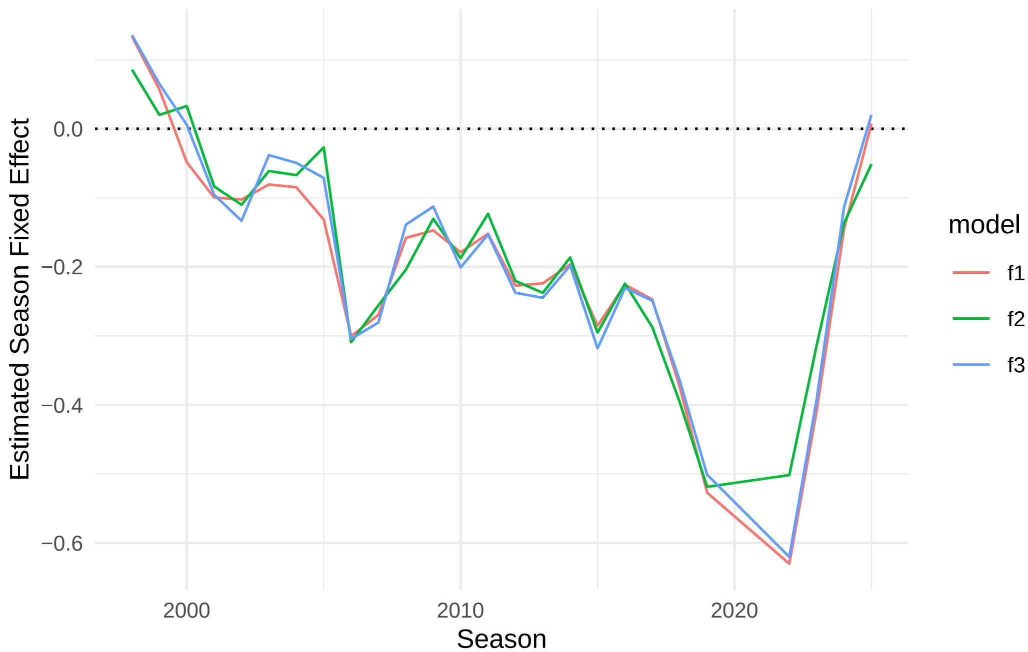

Figure 5: Estimated season-level fixed effects for natural logarithm of attendance models.

Source: Article Notebook

Figure 6 shows a general increase in the slope of the interaction terms in all three models in recent years. For all three models, these change in slope coefficients were decidedly negative until around 2015. This suggests that the relationship between attendance and fan engagement was more moderate during those years. (Note that for all three models, the main effect of the fan engagement term in Table 3 is positive.) While the change in slope associated with model f_1 is still negative, it is closer to zero than it has been in the past. The changes in slope associated with models f_2 and f_3 are now positive, although it is difficult to assess the long-term trend due to the pandemic being only a few years past.

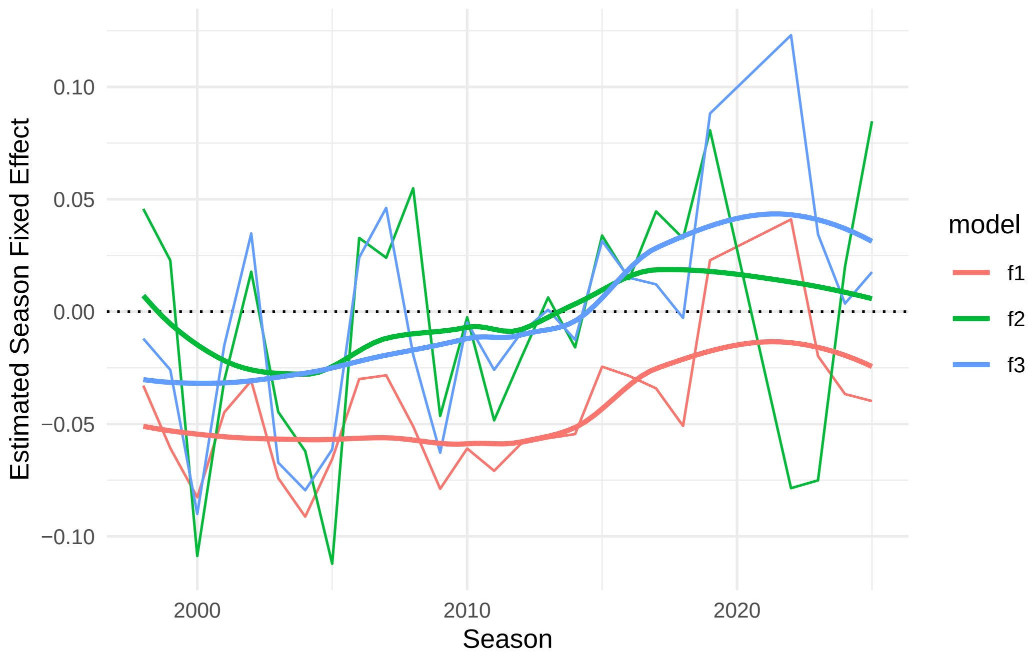

Figure 6: Estimated season-level changes in slopes for natural logarithm of attendance models.

Source: Article Notebook

4.2 Percentage of Arena Capacity

Because our response variable γ is constrained between 0 and 1, but has significant mass at 1 exactly, we fit a Papke-Wooldridge fractional response model (Papke and Wooldridge 1996) with HC1 sandwich estimators for the covariance matrix.

Source: Article Notebook

Table 6 reports the coefficients from the percentage of arena capacity models.

Table 6: Table of coefficients from percentage of arena capacity models.

model	term	estimate	std.error	statistic	p.value
f1	(Intercept)	0.0420351	0.3060837	0.1373320	0.8907684
f1	I(elo.A + elo.B - 3000)	0.0053038	0.0012483	4.2489239	0.0000215
f1	median_income	0.0192855	0.0046684	4.1310605	0.0000361
f1	median_pop	-0.0975090	0.0147117	-6.6279729	0.0000000
f1	is_cc	2.8268291	0.2623045	10.7768979	0.0000000
f2	(Intercept)	0.3066991	0.3737256	0.8206531	0.4118439

Table 6: Table of coefficients from percentage of arena capacity models.

model	term	estimate	std.error	statistic	p.value
f2	I(abs(elo.A - elo.B))	-0.0012487	0.0029263	-0.4267265	0.6695786
f2	median_income	0.0164644	0.0046623	3.5314191	0.0004133
f2	median_pop	-0.0948250	0.0144747	-6.5511069	0.0000000
f2	is_cc	2.9022797	0.2595776	11.1807804	0.0000000
f3	(Intercept)	0.0502670	0.3202932	0.1569407	0.8752916
f3	I(home_win_prob - 0.5)	2.3584200	0.8986874	2.6242941	0.0086829
f3	median_income	0.0179402	0.0047330	3.7904230	0.0001504
f3	median_pop	-0.1017634	0.0149716	-6.7970886	0.0000000
f3	is_cc	2.8717453	0.2611669	10.9958230	0.0000000

Source: Article Notebook

Table 7 shows the model fit metrics for the percentage of arena capacity models.

Table 7: Comparison of percentage of arena capacity models.

model	null.deviance	df.null	logLik	AIC	BIC	deviance	df.residual	nobs
f1	916.6665	5062	-2973.023	6104.046	6619.894	559.5177	4984	5063
f2	916.6665	5062	-2999.275	6156.550	6672.398	574.1421	4984	5063
f3	916.6665	5062	-2986.255	6130.510	6646.358	564.3610	4984	5063

Source: Article Notebook

Table 8 shows the analysis of variance for the percentage of arena capacity models.

Table 8: Analysis of variance for terms in models for percentage of arena capacity.

Model	Term	Df	Deviance	Df Residual	Residual Deviance	P Value
f2	I(abs(elo.A - elo.B))	1	0.032	5061	916.634	0.858
f1	I(elo.A + elo.B - 3000)	1	8.035	5061	908.631	0.005
f3	I(home_win_prob - 0.5)	1	0.906	5061	915.760	0.341
f1	is_cc	1	24.002	5010	573.411	0.000
f2	is_cc	1	23.389	5010	581.441	0.000

Table 8: Analysis of variance for terms in models for percentage of arena capacity.

Model	Term	Df	Deviance	Df Residual	Residual Deviance	P Value
f3	is_cc	1	22.520	5010	579.403	0.000
f1	median_income	1	0.977	5012	601.451	0.323
f2	median_income	1	1.038	5012	608.706	0.308
f3	median_income	1	0.972	5012	606.342	0.324
f1	median_pop	1	4.038	5011	597.414	0.044
f2	median_pop	1	3.876	5011	604.830	0.049
f3	median_pop	1	4.419	5011	601.923	0.036
f1	factor(month(game_date))	4	13.370	5025	696.118	0.010
f2	factor(month(game_date))	4	13.218	5025	706.988	0.010
f3	factor(month(game_date))	4	12.953	5025	701.429	0.012
f1	factor(wday(game_date))	6	18.624	5029	709.487	0.005
f2	factor(wday(game_date))	6	18.719	5029	720.205	0.005
f3	factor(wday(game_date))	6	19.034	5029	714.381	0.004
f1	name_current	12	93.689	5013	602.429	0.000
f2	name_current	12	97.243	5013	609.745	0.000
f3	name_current	12	94.115	5013	607.314	0.000
f2	I(abs(elo.A - elo.B)):factor(season_id)	26	7.299	4984	574.142	1.000
f1	I(elo.A + elo.B - 3000):factor(season_id)	26	13.894	4984	559.518	0.974
f3	I(home_win_prob - 0.5):factor(season_id)	26	15.042	4984	564.361	0.957
f1	factor(season_id)	26	180.519	5035	728.112	0.000
f2	factor(season_id)	26	177.710	5035	738.924	0.000
f3	factor(season_id)	26	182.345	5035	733.416	0.000

Source: Article Notebook

Figure 7 shows that…

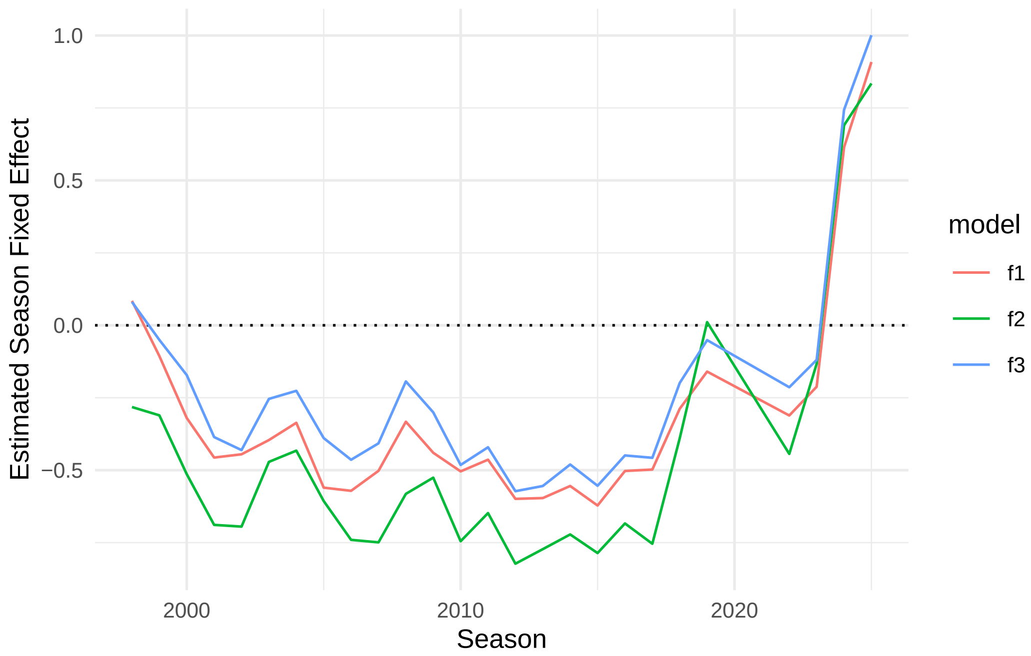

Figure 7: Estimated season-level fixed effects for percentage of arena capacity models.

Source: Article Notebook

Figure 8 shows that…

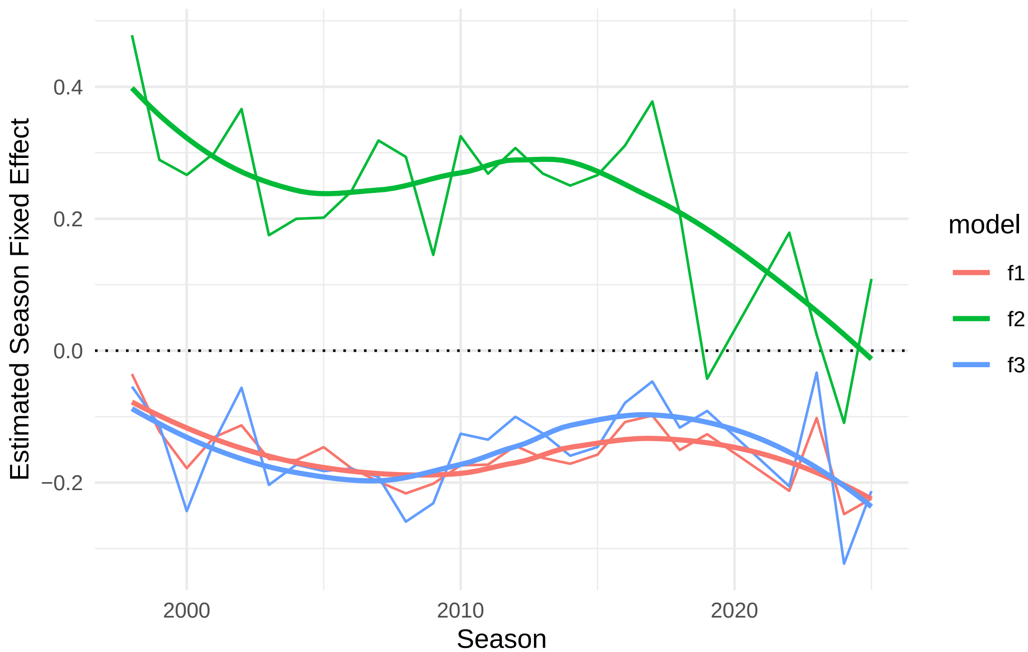

Figure 8: Estimated season-level changes in slopes for percentage of arena capacity models.

Source: Article Notebook

5 Conclusion

5.1 Limitations

Because the percentage of arena capacity data is right-censored, we considered a Tobit regression model (McDonald and Moffitt 1980), but our residuals violated the normality and homoskedasticity conditions. We also considered fitting a Beta regression model with a logit link function, and converting the fixed effects for seasons to random effects, but those models didn't converge.

5.2 Future work

5.3 Discussion

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6 Appendix

6.1 Residual analysis of attendance models

```
$f1
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$f2
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$f3
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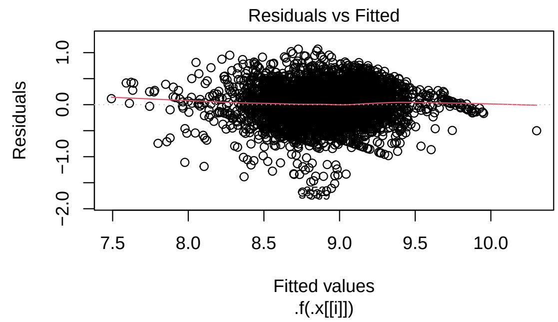
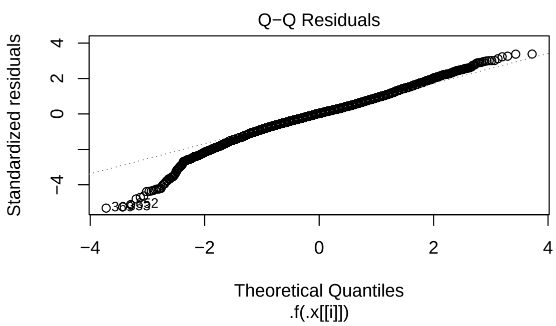


Figure 9: Residual plots for models.



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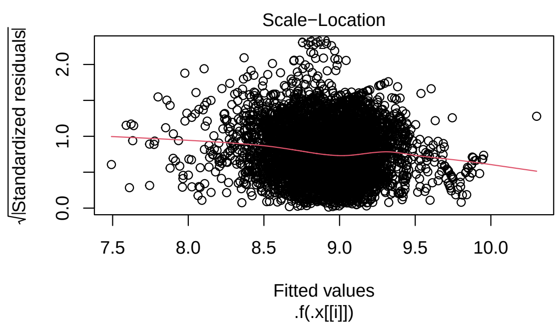

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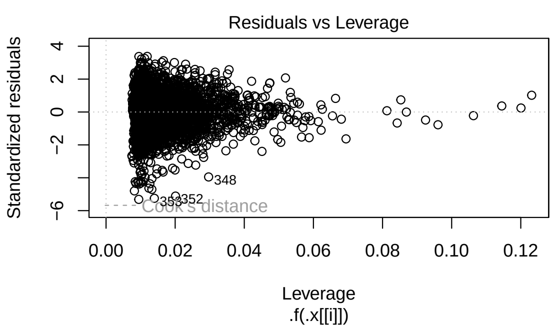

Figure 12: Residual plots for models.

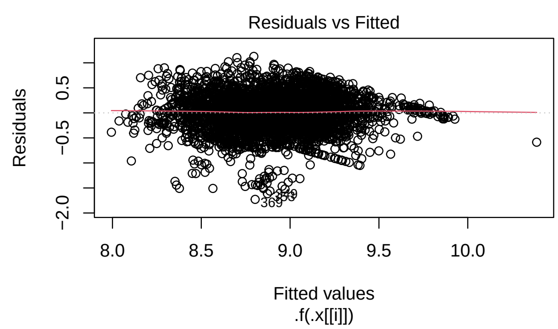

Figure 13: Residual plots for models.

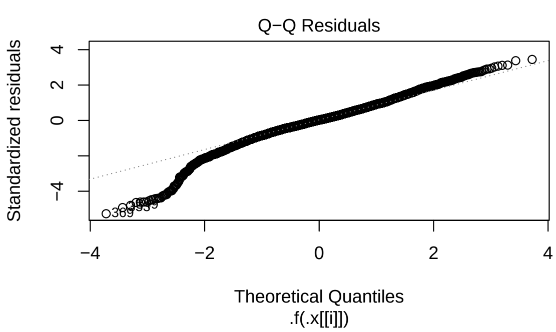

Figure 14: Residual plots for models.

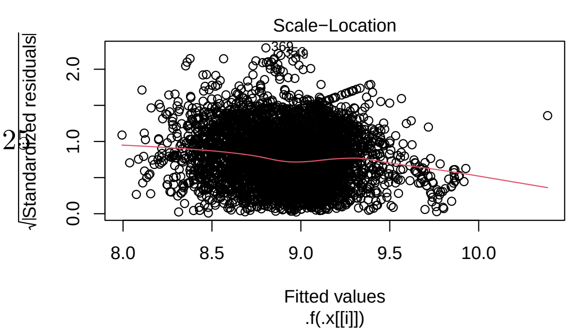

Figure 15: Residual plots for models.